THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — SCIENCE · NCERT-TYPE FULL-YEAR PAPER SET 1

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The balanced equation for burning of magnesium ribbon in air is: 1

(a) $2\,Mg + O_2 \longrightarrow 2\,MgO$ **(b)** $Mg + O_2 \longrightarrow MgO$ **(c)** $2\,Mg + O_2 \longrightarrow 2\,MgO_2$ **(d)** $Mg + O \longrightarrow MgO$

2. A solution turns red litmus blue. Its pH is likely to be: 1

(a) 10 **(b)** 4 **(c)** 7 **(d)** 1

3. Which of the following pairs will give a displacement reaction? 1

(a) $MgCl_2$ solution and aluminium metal **(b)** $Fe + CuSO_4$ solution

(c) $FeSO_4$ solution and silver metal **(d)** NaCl solution and copper metal

4. What would be the electron dot structure of carbon dioxide (CO_2)? The number of shared electron pairs in it is: 1

(a) 6 **(b)** 4 **(c)** 2 **(d)** 3

5. The kidneys in human beings are a part of the system for: 1

(a) nutrition **(b)** transportation of oxygen

(c) respiration **(d)** excretion

6. A part of the neuron where information is acquired is the: 1

(a) dendrite **(b)** axon **(c)** axon end **(d)** cell body

7. Binary fission in Amoeba differs from that in Leishmania because in Leishmania the division occurs 1

(a) at a definite plane related to the whip-like flagellum

(b) in the nucleus alone

(c) in any plane

(d) only after sporulation

8. Who is called the Father of Genetics? 1

(a) Gregor Johann Mendel **(b)** Robert Hooke

(c) Charles Darwin **(d)** Louis Pasteur

9. The angle of incidence is always equal to the angle of reflection. This is the 1

(a) law of refraction **(b)** second law of reflection

(c) first law of reflection **(d)** Snell's law

10. The image in the human eye is formed on the 1

(a) iris **(b)** pupil **(c)** retina **(d)** cornea

11. The SI unit of electric charge is 1

(a) ampere **(b)** coulomb **(c)** ohm **(d)** volt

12. The magnetic field lines outside a bar magnet are directed 1

(a) along the magnet only **(b)** from the south pole to the north pole

(c) in no definite direction **(d)** from the north pole to the south pole

13. An ecosystem consists of 1

(a) only animals **(b)** only plants

(c) biotic and abiotic components **(d)** only non-living things

14. Which of the following is a double displacement reaction? 1

(a) $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$ **(b)** $C + O_2 \longrightarrow CO_2$ **(c)** $Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2\,NaCl$ **(d)** $CaCO_3 \longrightarrow CaO + CO_2$

15. The pH of gastric juice is about: 1

(a) 1.4 **(b)** 7.4 **(c)** 5.5 **(d)** 10

16. The reactivity of sodium is explained by: 1

(a) its high melting point **(b)** its large mass
(c) its easy loss of an electron **(d)** its hardness

17. Soaps are sodium or potassium salts of: 1

(a) short-chain alcohols **(b)** alkanes
(c) aldehydes **(d)** long-chain carboxylic acids

18. Aerobic respiration in humans yields: 1

(a) lactic acid **(b)** ethanol and CO_2
(c) only energy **(d)** CO_2, water and energy

19. The movement of a shoot towards light is called: 1

(a) geotropism **(b)** phototropism **(c)** chemotropism **(d)** hydrotropism

20. The male gamete in a flowering plant is formed in the 1

(a) stigma **(b)** ovule **(c)** pollen grain **(d)** ovary

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. Why should a magnesium ribbon be cleaned before it is burnt in air? 2

22. Why does dry HCl gas not change the colour of dry litmus paper? 2

23. Give an example of a metal which 2
(i) is a liquid at room temperature and
(ii) can be easily cut with a knife.

24. Why does carbon form compounds mainly by covalent bonding? 2

25. What are the differences between autotrophic nutrition and heterotrophic nutrition? 2

26. What is the difference between a reflex action and walking? 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. Why does the colour of copper sulphate solution change when an iron nail is dipped in it? Write the equation. 3

28. Write the general reaction of an acid with a metal. Give the reaction of dilute HCl with zinc and the test for the gas. 3

29. A hammer, a battery, a bulb, wires and a switch are given. How will you distinguish between metals and non-metals using them? 3

30. Draw the structures of the two isomers of butane and name them. 3

31. Why is the process of respiration considered an exothermic process? Write its equation. 3

OR

Draw/describe the reflex arc for touching a hot object. 3

32. Describe the process of budding in Hydra. 3

OR

A pure tall plant is crossed with a pure dwarf plant. Write the genotype and phenotype of F1 and the F2 ratio. 3

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. What is a balanced chemical equation? Balance $Fe + H_2O \longrightarrow Fe_3O_4 + H_2$ step by step and write state symbols. 5

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34. Describe the reaction of dilute HCl with zinc granules. State the observations, the test for the gas, the equation and the general reaction. 5

35. Explain with equations the reactions of metals with oxygen, giving Na, Mg, Al and Cu as examples. State which oxide is amphoteric. 5

36. Explain the bonding in carbon: four valence electrons, covalent bond, catenation and tetravalency. Draw dot structure of methane. 5

OR

Describe the steps of photosynthesis: absorption of light, splitting of water, reduction of CO_2. What happens in desert plants at night? 5

37. Describe the structure of a neuron, with the path of a nerve impulse, and the synapse. 5

OR

Explain the different modes of asexual reproduction in single organisms with one example each. 5

38. Explain Mendel's monohybrid cross with tall and dwarf pea plants. Show the F1 and F2 generations with a Punnett square and the ratios. 5

OR

Describe the images formed by a concave mirror for different positions of the object (at infinity, beyond C, at C, between C and F, at F, between F and P) with nature and size. 5

END OF QUESTION PAPER **Total: 80 Marks**

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